

Lab 06 – Worksheet

Seven Segment Decoder Design

Name:	ID:	Section:
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Task a) Introduction – Add/Subtract Module

Label the figure below or type a test case in the box given below which you have performed on given design during the lab:

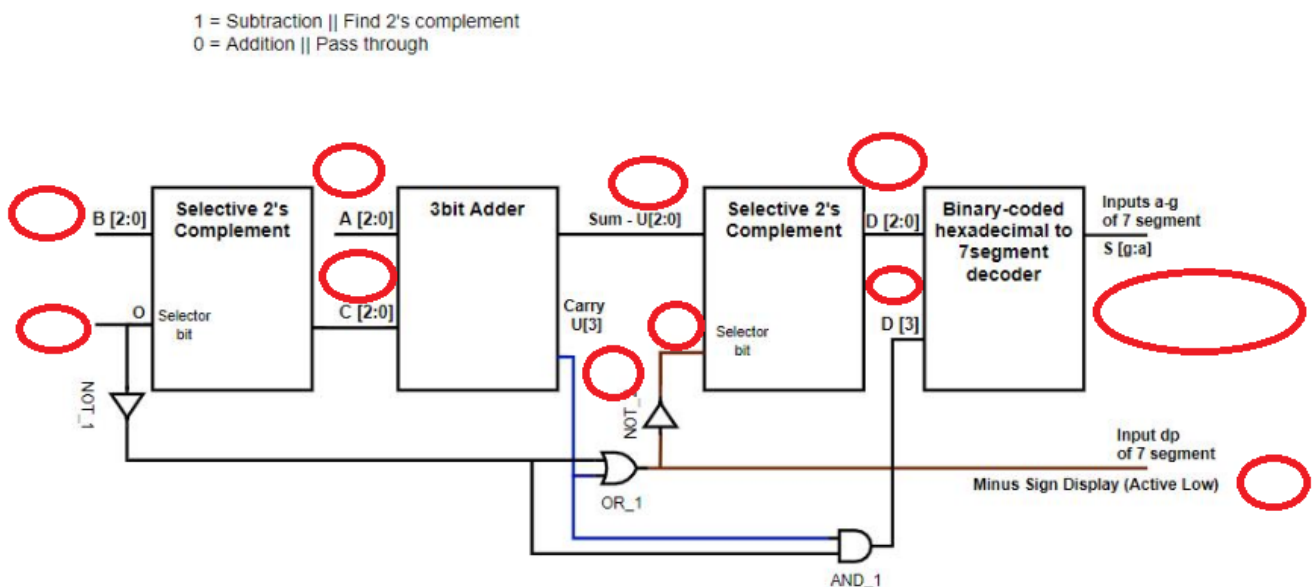


Figure 1: Add/Subtract Module

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Task b) 7-Segment Hexadecimal Display Decoder

(i) Complete the truth table of the Seven Segment Decoder in Table 1.

Hex Digit	Binary-coded Hexadecimal/ Inputs to decoder				Outputs of the decoder						
	D ₃	D ₂	D ₁	D ₀	S _g	S _f	S _e	S _d	S _c	S _b	S _a
0	0	0	0	0	1	0	0	0	0	0	0
1	0	0	0	1							
2											
3											
4											
5											
6											
7											
8											
9											
A											
B											
C											
D											
E											
F											

Table 1: Truth Table of Common-Anode seven segment display decoder

(ii) Write Boolean expression for each output segment in Table 2.

$S_a =$	
$S_b =$	
$S_c =$	
$S_d =$	
$S_e =$	
$S_f =$	
$S_g =$	

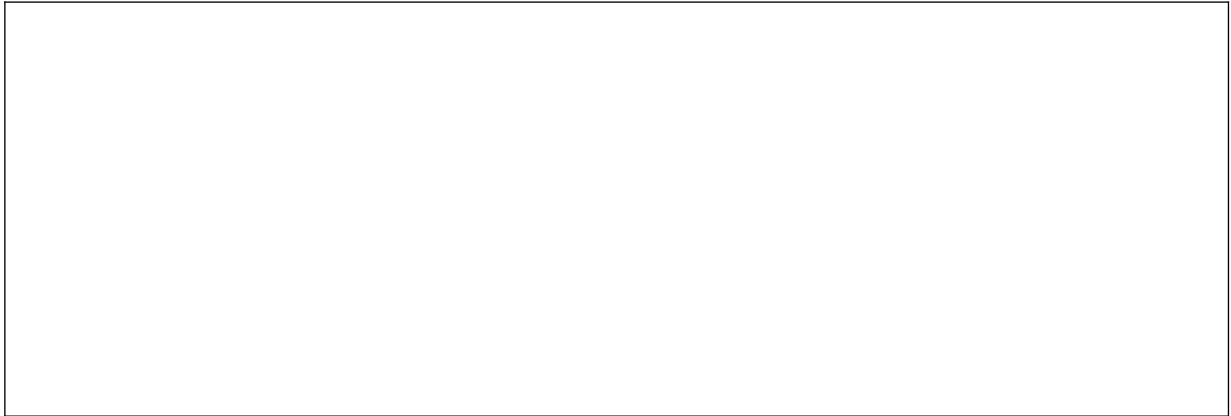
Table 2: Boolean expression for each output of seven segment decoder display

(iii) Appropriately add comments to your code and **write** the code of your design module in the box below.

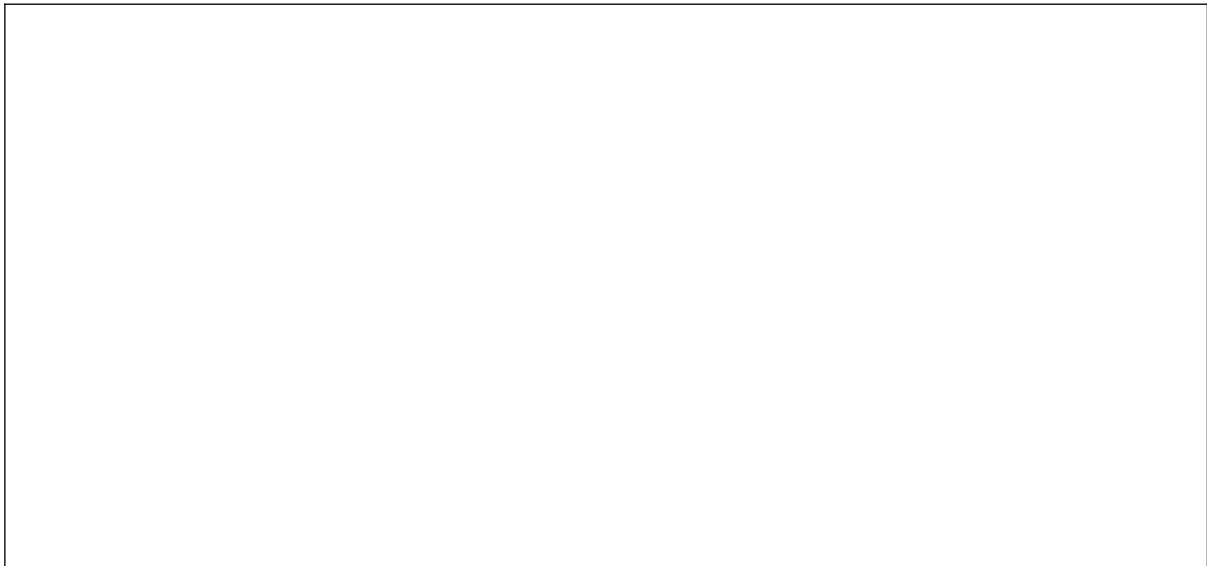
(iv) Add testbench code here.

(v) Attach RTL Schematic of the module.

(vi) Attach screenshot of your simulation output window: (Invert background color to white using settings option in simulation window)



(vii) Attach screenshot of log-window here.



Task c) Adder for 3-bit binary numbers

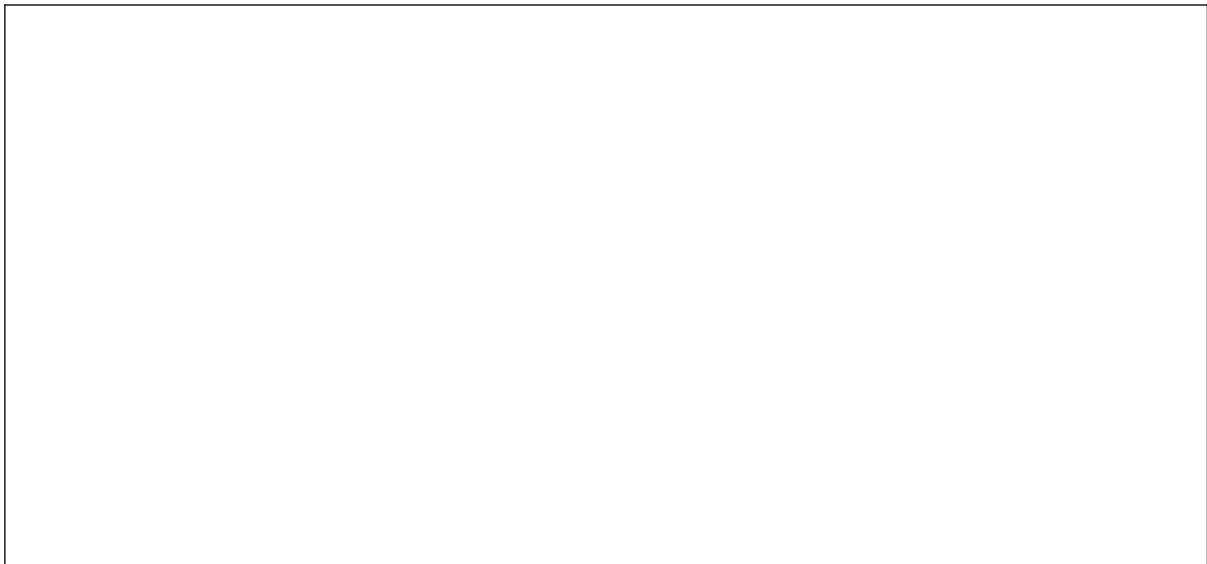
(i) Provide code for design module here.

(ii) Add testbench code here.

- (iii) Attach screenshot of waveform results here (Invert background color to white using settings option in simulation window)



- (iv) Attach screenshot of log-window here.



Assessment Rubric Lab 06 - Worksheet Seven Segment Decoder Design

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Marks Distribution:

		LR2 Code	LR4 Data Collection	LR5 Results	LR7 Viva	LR9 Report
In - Lab	Task a	-	-	-	10	10
	Task b	15	20	15		-
	Task c	20	-	10		-
Total Marks: 100		/35	/20	/25	/10	/10
CLO Mapping		CLO2	CLO2	CLO2	CLO4	CLO4

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	Viva	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR9	Report	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in- lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots and figures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plots and figure with proper label, titles and captions